

Proofs of Proposition and Theorems

The Graph Signature: A Query Optimization Index for RDF Graph Databases Using Bisimulation and Trace Equivalence

Mustafa Jarrar, Anton Deik

*Sina Institute, Birzeit University,
P.O.Box 14, Birzeit, West Bank, Palestine*
mjarrar@birzeit.edu, anton.deik@gmail.com

This document comes as an appendix to our paper entitled: ‘The Graph Signature: A Query Optimization Index for RDF Graph Databases Using Bisimulation and Trace Equivalence’. It presents the proofs of the proposition and theorems presented in the paper. Although our proofs can be shortened, for the sake of understandability, they are simplified and presented below in a semi-formal way.

Proof of Proposition. If we prove that every Trace Equivalence class is a partition set of some Bisimilarity classes, the proof becomes straight forward. It is proved by Milo and Suciu [1], that Bisimilarity implies Trace Equivalence: Let G be our data graph, $u, v \in G$, the following is true; if u, v are Bisimilar ($\approx$), then it is implied that they are Trace Equivalent ($\equiv$), formally: $u \approx v \Rightarrow u \equiv v$. This means that if some nodes $(v_1, v_2, v_3, \dots, v_n) \in G$ are Bisimilar, it is implied that they are Trace Equivalent: $(v_1 \approx v_2 \approx \dots \approx v_n) \Rightarrow (v_1 \equiv v_2 \equiv \dots \equiv v_n)$. It follows from this and from Definitions 5 and 5’ of the Graph Signature that, given a Bisimilarity Equivalence Class (B) of nodes $(v_1, v_2, v_3, \dots, v_n) \in G$, then it follows that there exists exactly one Trace Equivalence Class (T), such that: $B \subseteq T$. From this and from Definitions 5 and 5’, it follows that T consists of one or more Bisimilarity classes, that is: $T = B_1 \cup B_2 \cup \dots \cup B_n$. In addition, Bisimilarity classes are disjoint (by Definitions 5 and 5’ of the Graph Signature), that is; for any Bisimilarity classes B_i, B_j : $B_i \cap B_j = \Phi$, wherever $i \neq j$. As a consequence, every Trace Equivalence class is a partition set of some Bisimilarity classes. Therefore, if a query evaluated on the Trace Equivalence Summary of graph G retrieves a precise answer, then its answer, when evaluated on the Bisimilarity summary of the same G , is precise.

Proof of Theorem 1. An arbitrary query Q is in the most general case of a graph-shaped query, that is, it is a graph pattern to be matched against the original data graph G . If the graph pattern matches a sub-graph of G the answer will be the result set o_x . From the definitions of both the I and the O Signatures, one can infer that both signatures maintain the paths between the equivalent classes as they are present between the nodes pertaining to those equivalent classes. Formally, let $\{o_1 p_1 o_2 p_2 \dots p_{n-1} o_{n-1} p_n o_n\}$ be a query path and let p_1 be an edge from o_1 to o_2 in G , then (by definition) its O/I-Signature must contain p_1 from O_1 to O_2 , and so for p_n from o_{n-1} to o_n there is p_n from O_{n-1} to O_n (where $o_1 \in O_1 \dots o_n \in O_n$). Hence for every path $\langle (o_1, o_2, p_1), (o_2, o_3, p_2), \dots, (o_{n-1}, o_n, p_{n-1}) \rangle$ in G there is exactly the same path in any of the I/O Signatures $\langle (O_1, O_2, p_1), (O_2, O_3, p_2), \dots, (O_{n-1}, O_n, p_{n-1}) \rangle$. So, if the graph pattern matches a sub-graph of G , then it matches the corresponding sub-graph in any of the signatures, resulting in the result set O_x . And because the result set O_x consists of equivalent classes of the nodes in o_x , then O_x is a superset of o_x (i.e., $o_x \subseteq O_x$).

Proof of Theorem 2. Because (as proven in theorem 1) the answer of the I/O-Signature is safe, i.e., a superset of the target answer, then if this superset is empty, the target answer (subset of the answer of the I/O-Signatures) is empty.

Proof of Theorem 3. As argued in the proof of Theorem 1, for every path $\langle (o_1, o_2, p_1), (o_2, o_3, p_2), \dots, (o_{n-1}, o_n, p_{n-1}) \rangle$ in G there is exactly the same path in any of the I/O Signatures $\langle (O_1, O_2, p_1), (O_2, O_3, p_2), \dots, (O_{n-1}, O_n, p_{n-1}) \rangle$, where

$o_1 \in O_1 \dots o_n \in O_n$. Given that all query nodes are unbound, this implies that the edge paths retrieved from any of the I/O Signatures are equal to the paths retrieved from the original data graph G .

Proof of Theorem 4. Query $?p_1, \dots, ?p_{n-1} : \{ \langle o_1 \rangle p_1 ?o_2 p_2 \dots p_{n-1} ?o_n \}$ asks for paths outgoing from node o_1 , given that all other nodes are unbound. The definition of the O-Signature implies that, for a node $o_1 \in G$ and its equivalence class O_1 in the O-Signature, the set of all outgoing paths from o_1 is equal to the set of all outgoing paths from O_1 and it equals the set of all outgoing paths from any node x in O_1 . This implies that the O-Signature answers this query precisely.

Proof of Theorem 5. This is the inverse of the proof of Theorem 4. Query $?p_1, \dots, ?p_{n-1} : \{ ?o_1 p_1 ?o_2 p_2 \dots p_{n-1} \langle o_n \rangle \}$ is asking for paths incoming into node o_n , given that all other nodes are unbound. The definition of the I-Signature implies that for a node $o_n \in G$, and its equivalence class O_n in the I-Signature, the set of all incoming paths into o_n is equal to the set of all incoming paths into O_n and it equals the set of all incoming paths into any node x in O_n . This implies that the I-Signature answers this query precisely.

Proof of Theorem 6. Query $?p_1, \dots, ?p_{n-1} : \{ ?o_1 p_1 \dots \langle o_i \rangle \dots p_{n-1} ?o_n \}$, where $1 \leq i \leq n$, asks for all paths passing through node o_i and all other nodes are unbound. Given node $o_i \in G$ and its equivalence class in the O-Signature (O_i), by the definition of the O-Signature, all paths outgoing from o_i are equal to all paths outgoing from O_i (as described in the proof of Theorem 4). Also, if a path is incoming into o_i then it is incoming into O_i . However, O_i might have paths incoming into it that are not incoming into o_i . This means that the answer of S_O might include false positives which are false incoming paths joined with precise outgoing paths. The inverse is true for the I-Signature; its false positives are precise incoming paths joined with false outgoing paths. Hence, intersecting both answers eliminates the false positives, resulting in precise answers.

Proof of Theorem 7. Query $?o_1 : \{ ?o_1 p_1 ?o_2 p_2 \dots p_{n-1} ?o_n \}$ asks for all nodes that have a certain set of paths originating from them, given that all query nodes are unbound. The definition of the O-Signature implies that, for an equivalence class O in the O-Signature of the data graph G ; every node of G grouped in O have the same set of outgoing paths, which is equal to the set of outgoing paths from O . This implies that the results of executing the query on the O-Signature are Equivalence Classes that only contain all nodes having the required set of paths originating from them (i.e., the nodes of target answer), which means that this result set is precise.

Proof of Theorem 8. This is the inverse of the proof of Theorem 8. Query $?o_n : \{ ?o_1 p_1 ?o_2 p_2 \dots p_{n-1} ?o_n \}$ asks for all nodes that have a certain set of paths terminating into them, given that all query nodes are unbound. The definition of the I-Signature implies that, for an equivalence class O in the I-Signature of the data graph G ; all nodes of G that are grouped in O have the same set of incoming paths, which is equal to the set of incoming paths into O . This implies that the results of executing the query on the I-Signature are Equivalence Classes that only contain all nodes that have the required set of paths terminating into them (i.e., the nodes of target answer), which means that this result set is precise.

Proof of Theorem 9. Query $?o_i : \{ ?o_1 p_1 \dots ?o_i \dots p_{n-1} ?o_n \}$ (where $1 \leq i \leq n$), asks for node(s) o_i with a certain set of paths passing through them, given that all query nodes are unbound. As argued previously, the definition of the O-Signature implies that, for an equivalence class O in the O-Signature of the data graph G ; all nodes of G that are grouped in O have the same set of outgoing paths, which is equal to the set of outgoing paths from O . Also, if a path is incoming into any node in O then it is incoming into O in the O-Signature. However, the paths incoming into the nodes of O might not be equal. This means that the answer of S_O might include false positives which are the nodes that have precise outgoing paths but false incoming paths. The inverse is true for the I-Signature; its false positives are the nodes that have precise incoming paths but false outgoing paths. Hence intersecting both answers eliminates the false positives, resulting in precise answers.

References

1. Milo, T., and Suciu, D.: Index structures for path expressions. ICDT 1999.